

Applied Thermodynamic II

ME2402

Refrigeration & Air Conditioning



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Content

- Reversed Heat Engine
- Ideal cycle and Vapour Compression Refrigeration
- Refrigerating load
- Compressor work and Coefficient of performance
- Modifications to the ideal cycle
- Use of flash chamber
- Specific Humidity, Relative Humidity and Percentage Saturation,
- Cooling Towers

1.0 Reversed Heat Engine

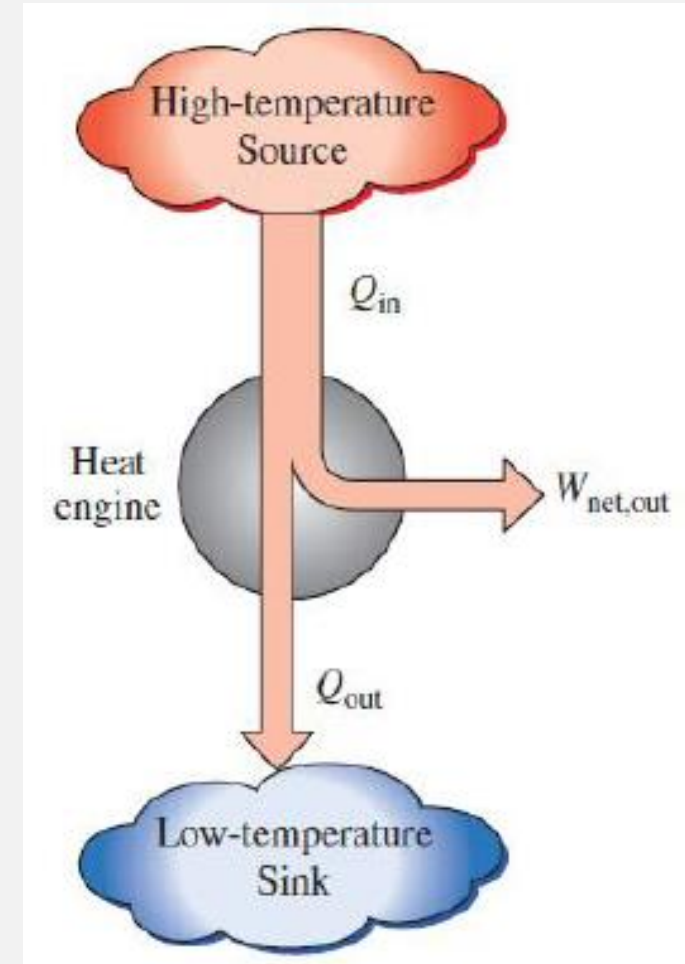


Natural Heat Transfer

- Heat naturally flow from high temperature body to low temperature body

Heat Engines

Convert part of the heat receiving from a high temperature body (source) to work and reject remaining to a low temperature body (sink)

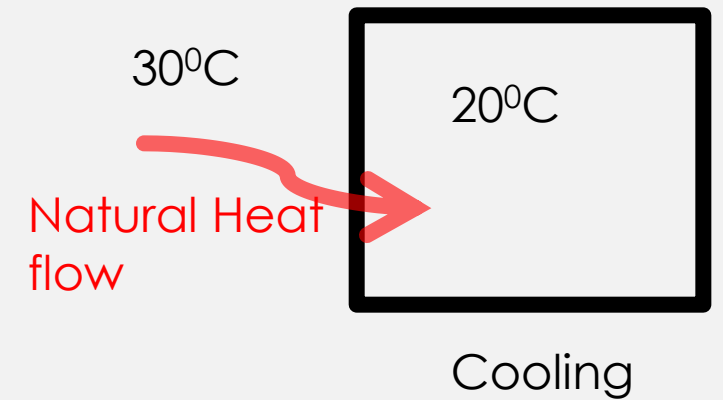


1.0 Reversed Heat Engine

Refrigeration and air conditioning

Cooling a space

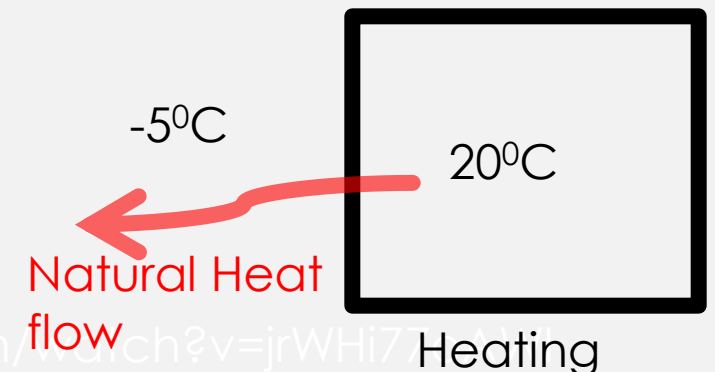
- Need **removing heat** from **low temperature** body to **high temp** surrounding
- Use refrigerators and air conditioners



Heating a space

- Need **sending heat** to a **high temperature** space from **low temp** surrounding
- Use heat pumps

Both of above **cannot** happen naturally. (2nd law)



1.0 Reversed Heat Engine

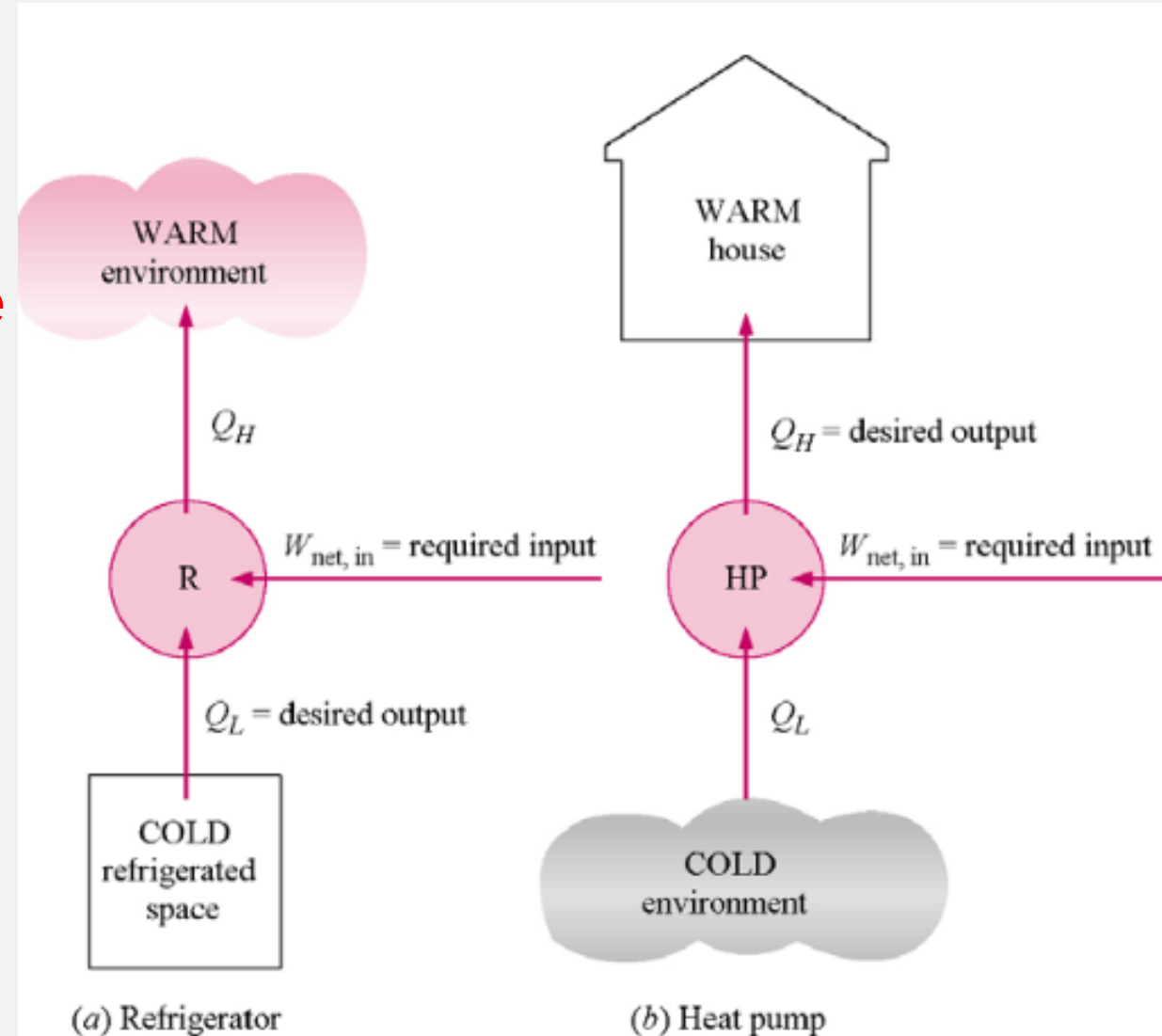
Refrigeration and air conditioning

Cooling a space

- Need removing heat from low temperature body to high temp surrounding
- Use refrigerators and air conditioners

Heating a space

- Need sending heat to a high temperature space from low temp surrounding
- Use heat pumps



1.0 Reversed Heat Engine



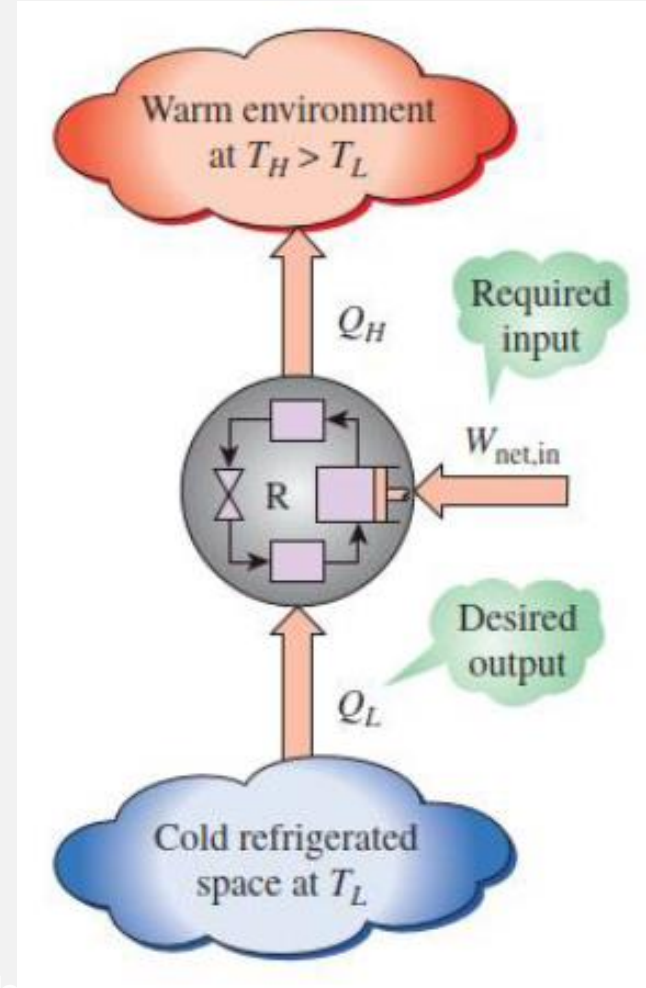
Forced Heat Transfer

- Heat **can be transfer** from **High** temperature body to **low** temperature body by **providing external work**.
(2nd law, Clausius statement)

Reversed Heat Engines

Reversed heat engine perform the above **Force Heat Transfer** by providing **external work** in to the system.

- Refrigerators and heat pumps operate on this cycle



1.0 Reversed Heat Engine



Reversed Heat Engines

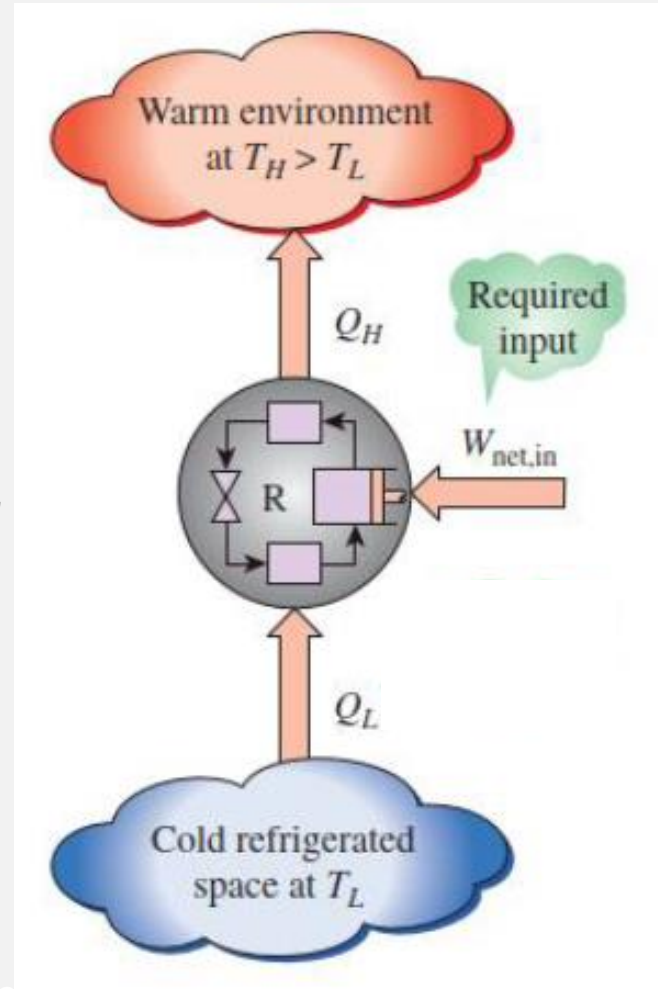
This is used in two different applications

- i. Cooling application : Refrigerators
 - Deliver heat to a space for heating
- ii. Heating application : Heat pumps
 - Remove heat from a space for cooling

Performance of reversed Heat Engine

Performance is evaluated using the value called Coefficient of performance (CoP)

$$CoP = \frac{\text{Desired Out put}}{\text{Net Work Input}}$$



1.0 Reversed Heat Engine

Performance of reversed Heat Engine (CoP)



i. Coefficient of performance Refrigerators

$$CoP_{Ref} = \frac{\text{Desired Out put}}{\text{Net Work Input}} = \frac{Q_L}{W}$$

ii. Coefficient of performance Heat pumps

$$CoP_{HP} = \frac{\text{Desired Out put}}{\text{Net Work Input}} = \frac{Q_H}{W}$$

From 2nd law,

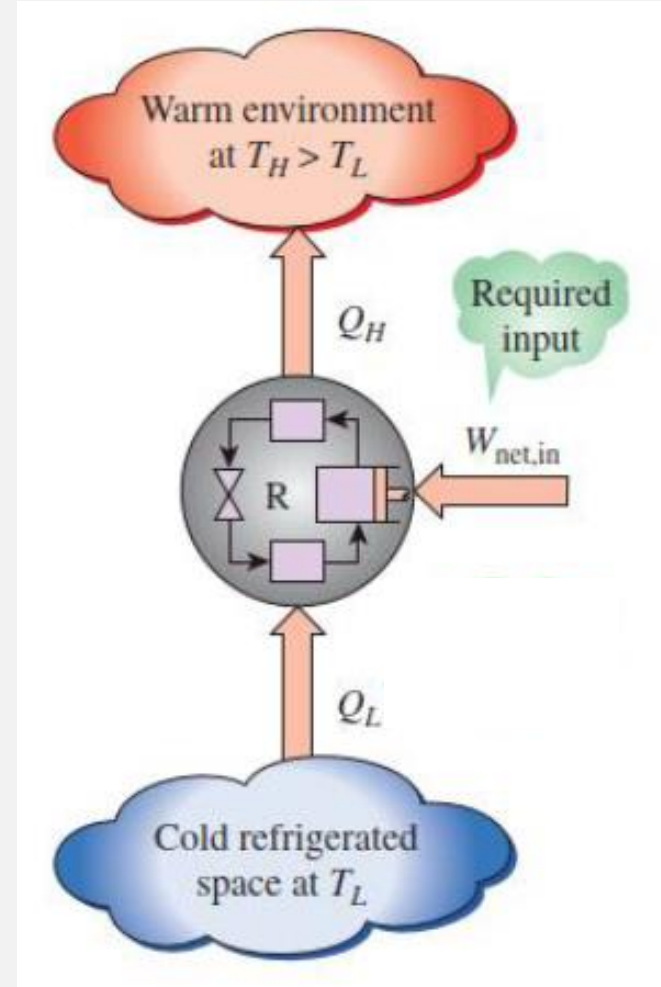
$$\sum dQ + \sum dW = 0$$

$$-Q_H + Q_L + W = 0 \quad \text{multiply by } 1/W$$

$$Q_L / W + W/W = Q_H / W$$



$$CoP_{Ref} + 1 = Cop_{Hp}$$

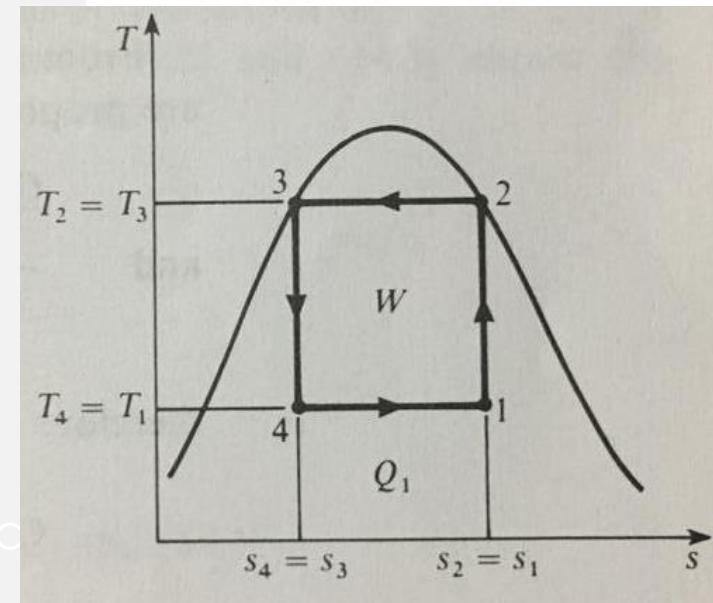
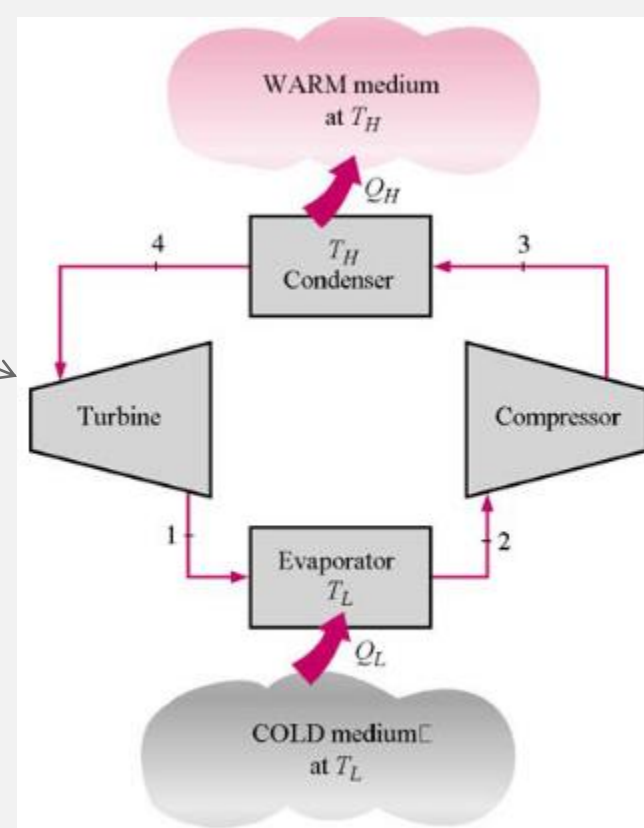


2.0 Reversed Carnot Refrigerator and Heat Pump

The **Best CoP** (maximum COP) is given by the **reversed Carnot cycle** operation

- This is a **theoretical cycle** (Ideal cycle) operating between **two temperature levels**, with maximum CoP

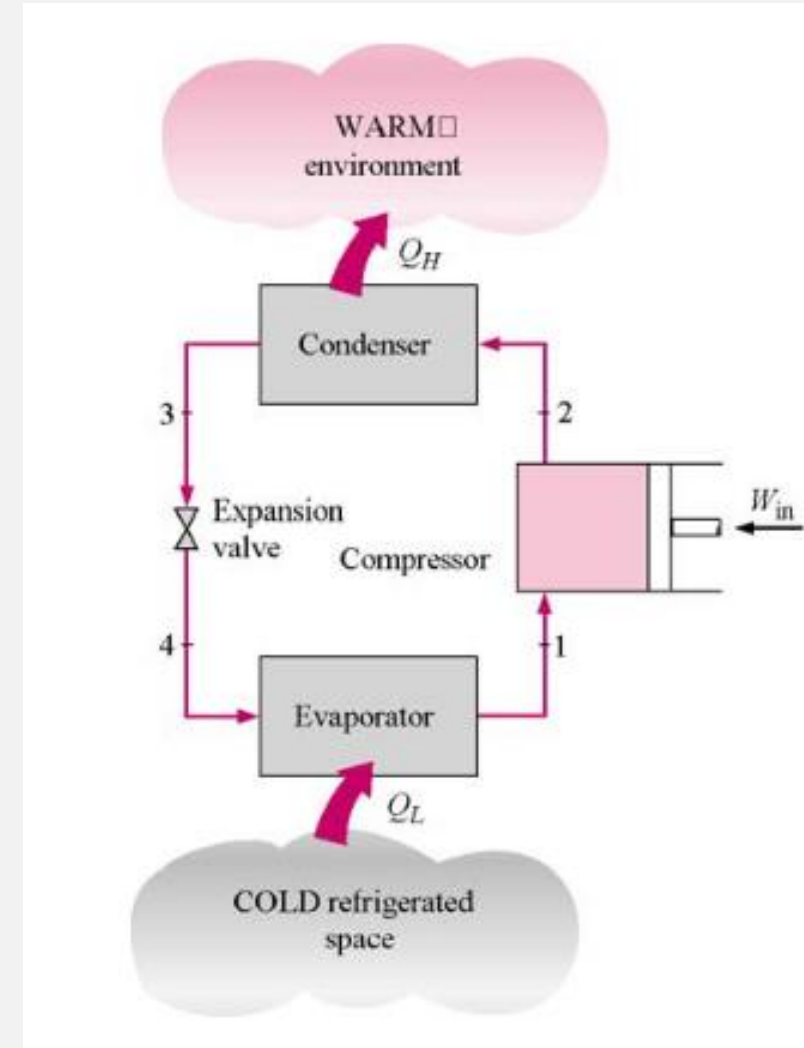
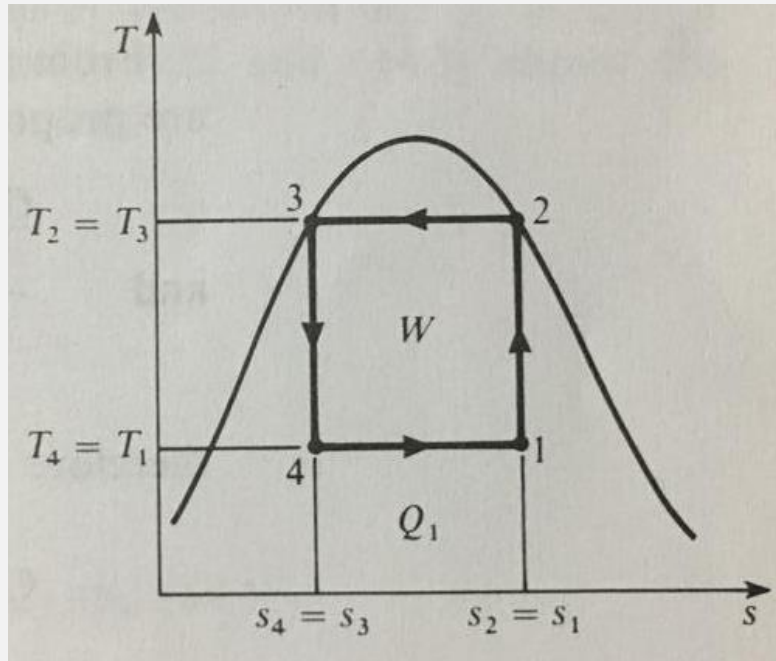
Expansion
engine



3.0 Vapor Compression Cycle

This is the widely used type of refrigerator type.

- Mainly consists of Evaporator, compressor, condenser and expansion device
- Evaporation and condensation occurs while refrigerant absorbing heat and rejecting heat
- Ideal cycle is as follows.



3.0 Vapor Compression Cycle

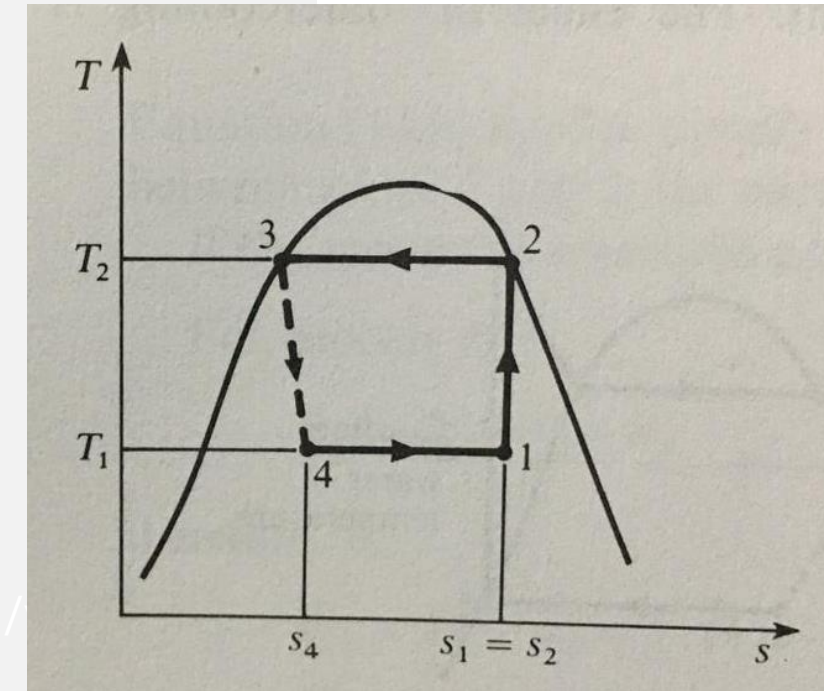
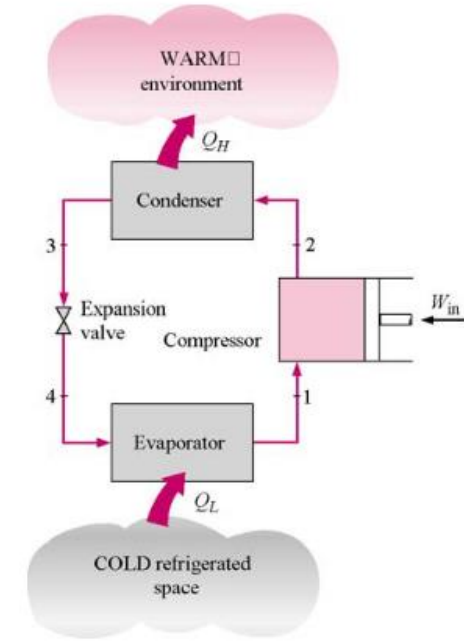
Several modifications are done for the ideal cycle as follows;

Modifications to ideal cycle

3.1.1 Use of expansion device

Now the cycle is having a **expansion device** instead of expansion engine

- Typically, a throttle valve
- This process is highly **irreversible** and given by dotted 3-4 line



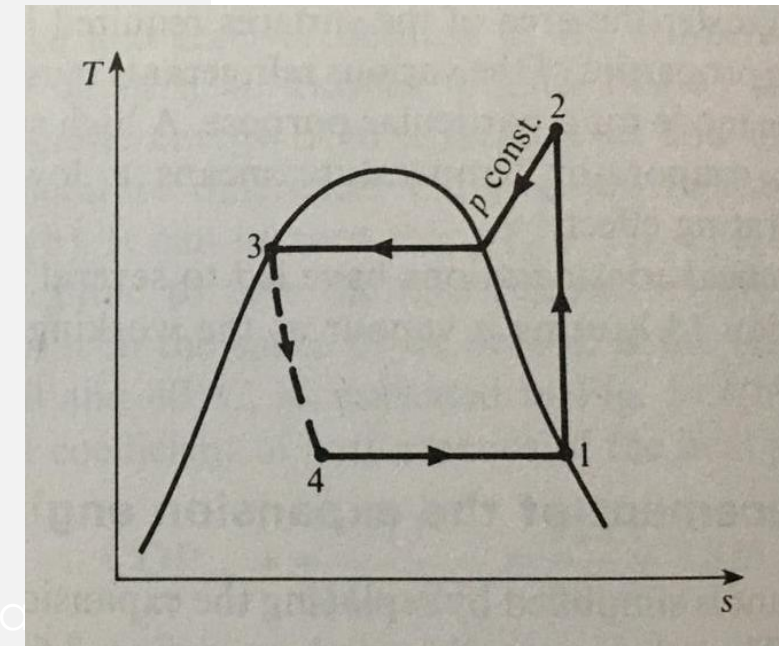
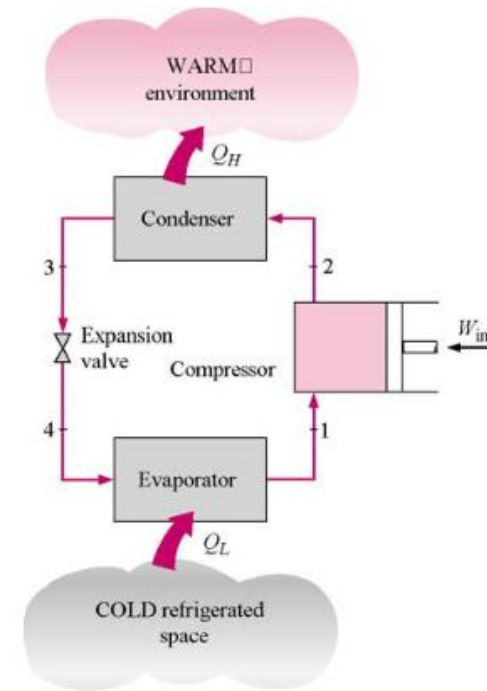
3.0 Vapor Compression Cycle

3.1.2 Conditioning at the compressor inlet

To use maximum specific enthalpy;

- Evaporation is continues only until vapor is **dry saturated (point 1- theatrically)**
- Practically, the vapor is **superheated** as it leaves evaporator
- Practically, **superheating** is essential to **avoid carry-over liquid** refrigerant to compressor
- If liquid enters, lubrications fails and damage compressor
- Superheat kept as much as **minimum**
- So the **point 2 goes to superheat region** and condensation (2-3) is **not constant temperature anymore**.

<https://www.youtube.com/watch?v=...>

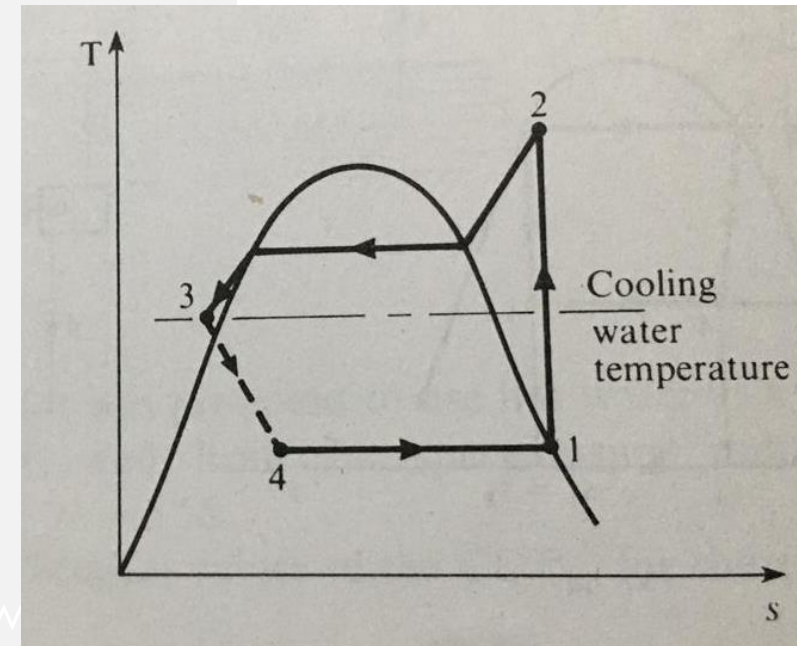
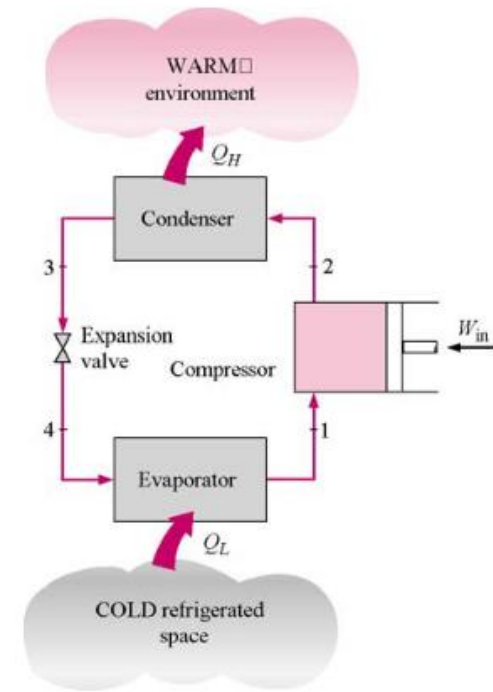


3.0 Vapor Compression Cycle

3.1.3 Undercooling of condensed vapor

To **increase the refrigeration effect** of process 4-1, we can shift 3-4 process to left in diagram. So,

- The condensed vapor is **further cooled below saturation temperature** at constant pressure.
- So the new throttling process (3-4) **more to left**
- However, the amount of under cooling is limited by the cooling water temperature.



3.0 Vapor Compression Cycle

3.2 Refrigerating Load

The amount of heat transferred per unit time from the cold chamber is known as refrigeration capacity.

- American unit for refrigeration is **ton**
- The **British thermal unit (BTU)** is a unit of heat and defined as the amount of heat required to raise the temperature of one pound of water by one degree Fahrenheit

The refrigeration capacity defines the mass flowrate of a refrigerant under the condition.

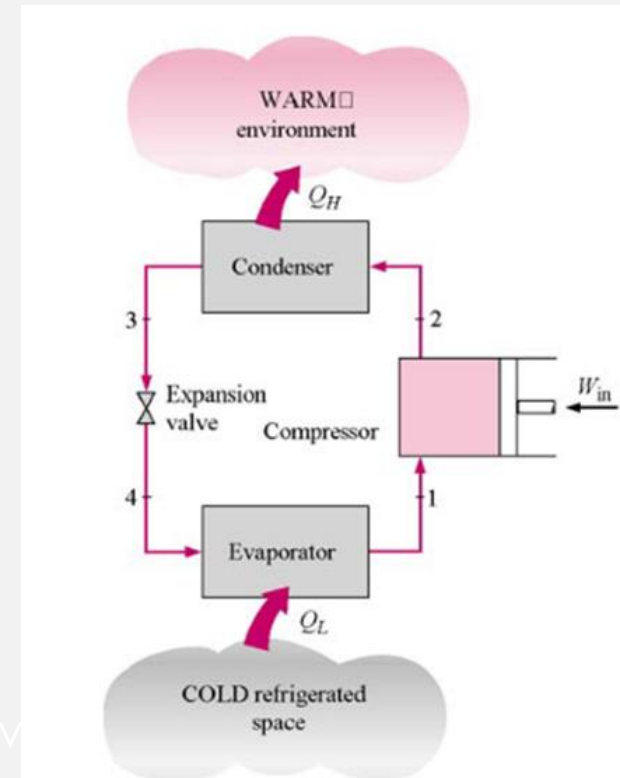
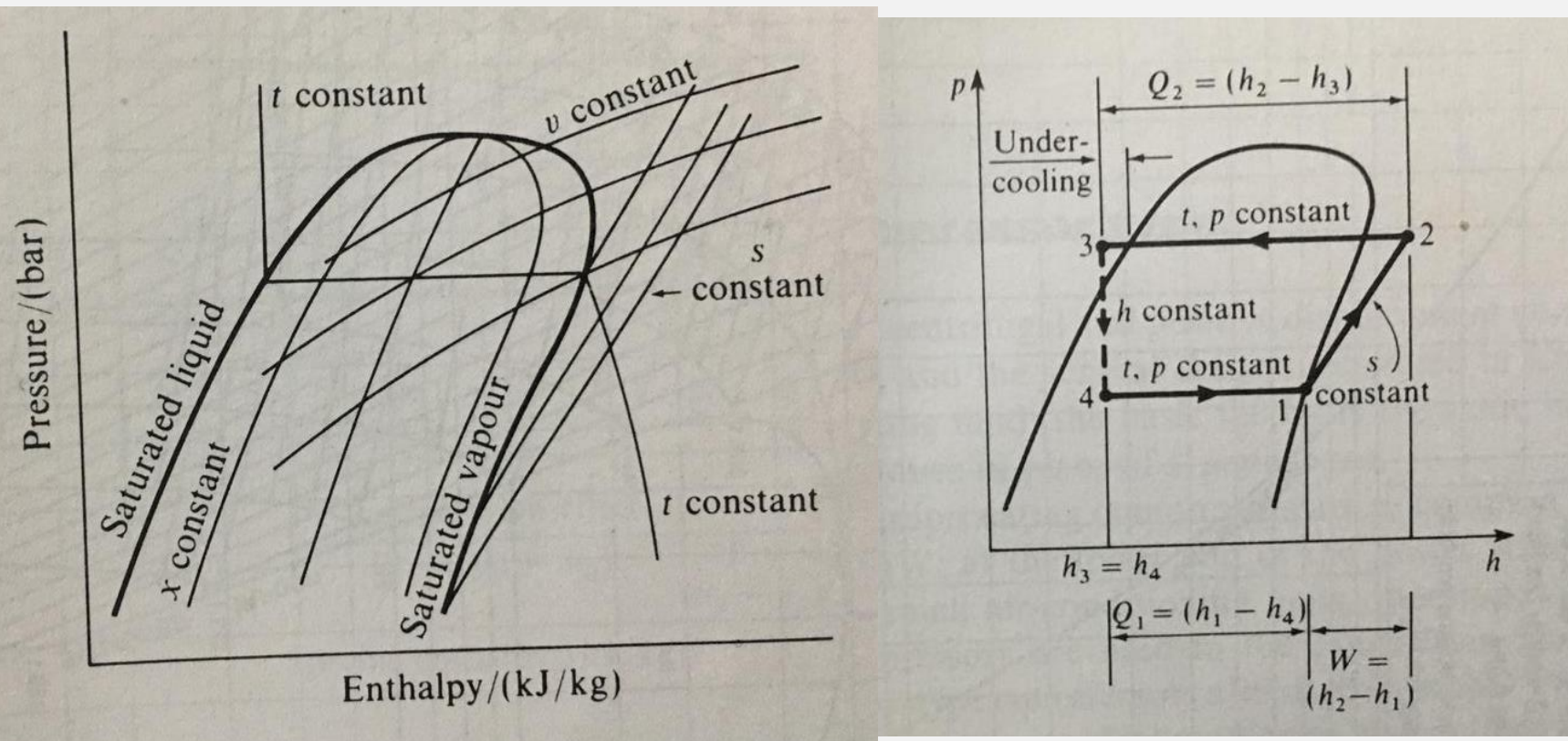
$$\text{mass flow rate}(\dot{m}) = \frac{\text{refrigeration capacity}}{\text{refrigeration effect per unit mass}}$$

3.0 Vapor Compression Cycle

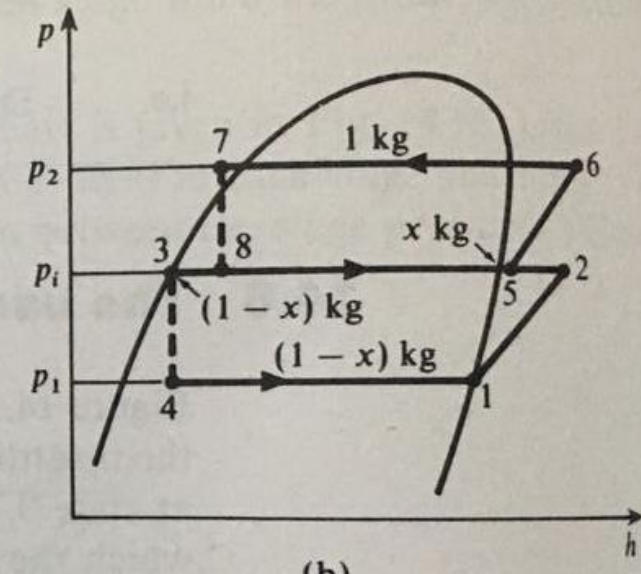
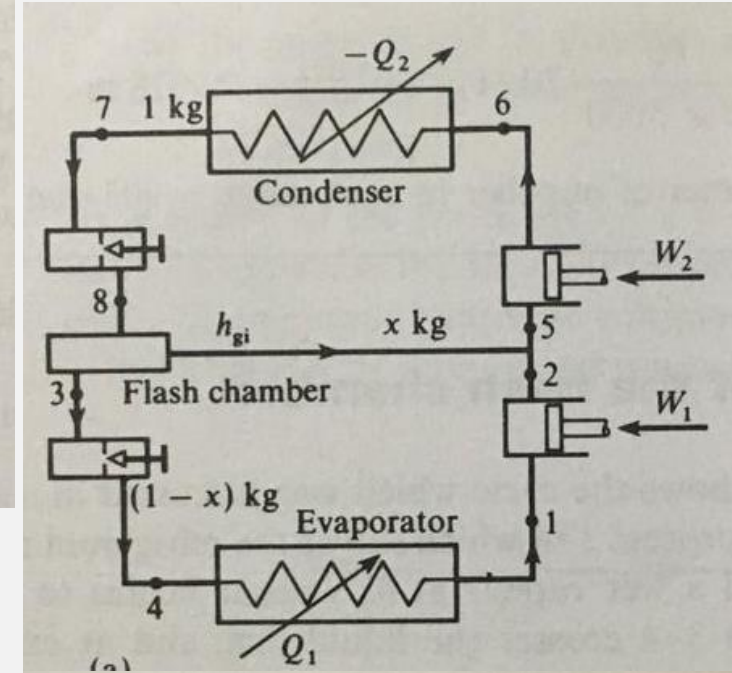
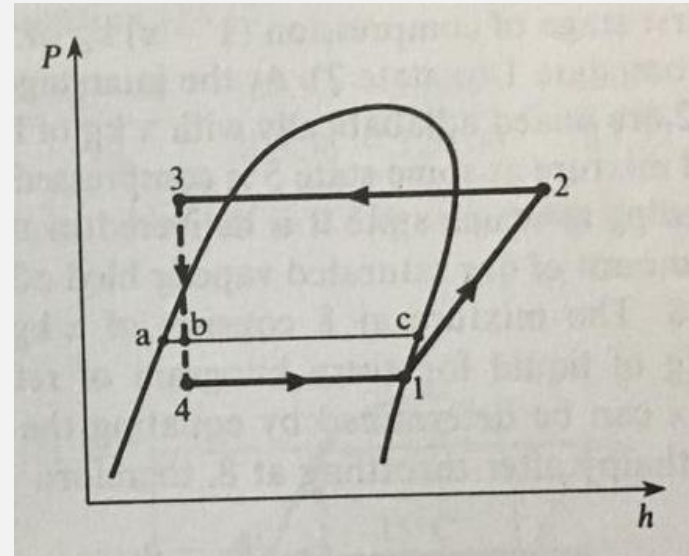
3.3 Pressure Enthalpy Diagram

This graph is essential for calculations

- In the diagram undercooling is visible



3.0 Vapor Compression Cycle



3.4 Use of Flash Chamber

- Process 3-4 is the **expansion** of **undercooled liquid** to wet vapor at 4.
- **Liquid refrigerant becoming vapor during** expansion and through valves is called **flashing**
- 4 contains vapor and liquid **but** vapor doesn't contribute for cooling
- So using a intermediate flash chamber this **vapor is bleed off** to compression process